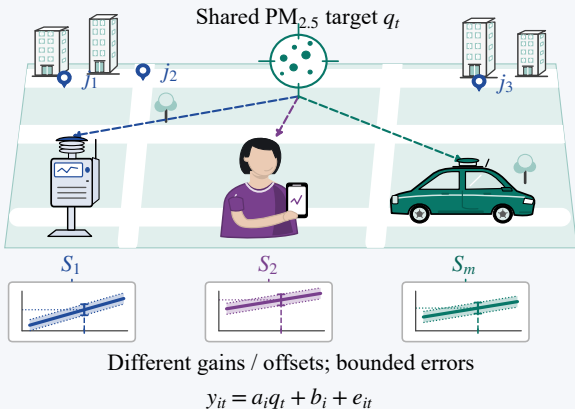


### (a) One target, heterogeneous sources



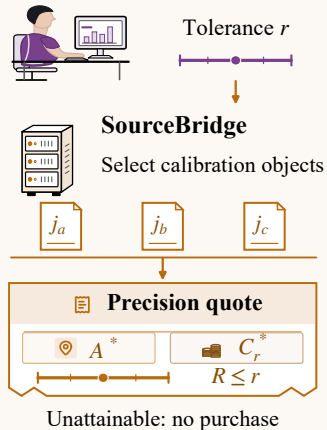
### (b) Public catalog

| Object | $I_j$ | Fee   |
|--------|-------|-------|
| $j_1$  |       | $c_1$ |
| $j_2$  |       | $c_2$ |
| $j_3$  |       | $c_3$ |
| $j_4$  |       | $c_4$ |

Available before purchase

|  |                              |
|--|------------------------------|
|  | $T, [a_{\min}, a_{\max}]$    |
|  | Error bounds $\varepsilon_j$ |

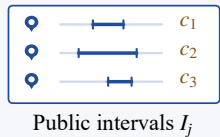
### (c) Precision contract



## Before collection

Public  $T$ ,  $[a_{\min}, a_{\max}]$ ,  $\varepsilon_j$ ;  $m > 2f$ ; request  $r$

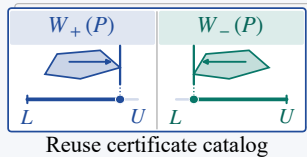
### 1 Public directory



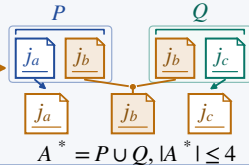
### 2 Supports



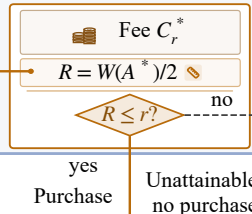
### 3 Two directional LPs



### 4 Cost union



### 5 Precision quote



Quoted radius  $R$  for projection

## After collection

### 7 Parallel inference

#### Nominal: calibrated estimator



#### Depth $\geq m - f$



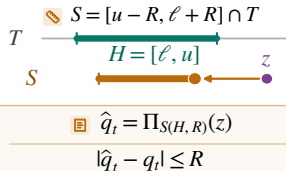
### 6 Complete packets

| $Y$   | $j_a$ | $j_b$ | $j_c$ | $t$ |
|-------|-------|-------|-------|-----|
| $S_1$ | $y$   | $y$   | $y$   | $y$ |
| $S_2$ | $y$   | $y$   | $y$   | $y$ |
| $S_3$ | *     | *     | *     | *   |
| $S_m$ | $y$   | $y$   | $y$   | $y$ |

$m \times (|A^*| + 1)$

At most  $f$  entire rows controlled

### 8 Safe projection



# 1 Catalog: $|P| \leq 2$



$n = 3$  example

$\emptyset$

1

2

3

1,2

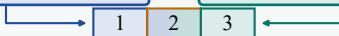
1,3

2,3

## 2 Directional certificates

$$P = \{1, 2\} \quad W_+(P) \leq \tau$$

$$Q = \{2, 3\} \quad W_-(Q) \leq \tau$$



## 3 Purchase $A = P \cup Q$



Charge once

Public

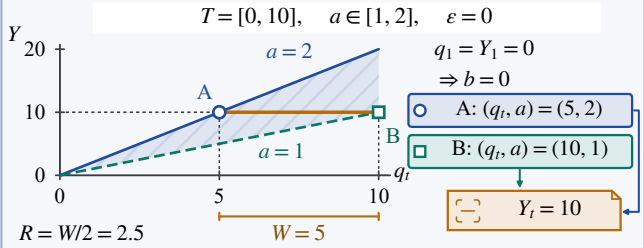
Reports

Cost split

|            | $j_1$                            | $j_2$ (shared)                   | $j_3$                            |
|------------|----------------------------------|----------------------------------|----------------------------------|
| Public     | $(I_1, \varepsilon_1)$           | $(I_2, \varepsilon_2)$           | $(I_3, \varepsilon_3)$           |
| Reports    | $Y_{11}$<br>$\vdots$<br>$Y_{m1}$ | $Y_{12}$<br>$\vdots$<br>$Y_{m2}$ | $Y_{13}$<br>$\vdots$<br>$Y_{m3}$ |
| Cost split | $j_1 : 3$                        | $j_2 : 2$                        | $j_3 : 4$                        |

$$C(P \cup Q) - C_0 = 3 + 2 + 4 = 9$$

## (a) Same-report ambiguity



## (b) Safe projection

